Registering LiDAR pointclouds for collaborative perception in autonomous driving

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***Abstract*—Cooperative perception allows vehicles to extend their perceptual range by sharing sensor data with roadside infrastructure, but fusing that data requires recovering the rigid transformation between the vehicle and infrastructure sensors. This is a point cloud registration problem, but one that differs substantially from the scan-to-scan setting most methods target, involving large viewpoint separation, limited and variable overlap, and an initial estimate that aggregates the localization errors of both agents. This work evaluates Voxelized Generalized ICP (VGICP), a classical, feature-free method, in this vehicle-to-infrastructure setting on the V2X-Real dataset, where it proves usable only when initialized close to the ground-truth transformation, with accuracy further conditioned by scene overlap. Since initialization rather than the method is the limiting factor, we characterize it with a controlled noise model perturbing each degree of freedom independently, finding registration far more sensitive to planar displacement and heading error than to roll, pitch or height displacement. To reduce the reliance on these noisy positional readings, we propose a sequential initialization strategy that derives each frame's estimate from the previous frame's registration, letting a single accurate estimate replace continuous positioning input over a bounded horizon before drift degrades it. Improving the registration itself proves harder: aside from a tuned voxel resolution, discarding height or ground geometry only degrades it. Findings are validated on real-world data without ground-truth poses.**

***Index Terms*—Point cloud registration, cooperative perception, vehicle-to-infrastructure, VGICP, sensor noise, pose initialization, autonomous driving, V2X-Real**

I. Introduction

COOPERATIVE perception has emerged as a key enabler for autonomous driving, allowing vehicles to extend their perceptual range and robustness by sharing sensor data with other vehicles and roadside infrastructure. Among the different collaboration modes, vehicle-to-infrastructure (V2I) cooperation is particularly attractive: infrastructure sensors are typically mounted at elevated, fixed positions, providing a stable, wide-coverage viewpoint that is largely unaffected by the vehicle's own occlusions and blind spots. The benefits of V2I cooperation are greater in low- or mid-level schemes that fuse sensor data or intermediate features. However, these strategies require that data from both agents be expressed in a common reference frame, which in turn requires an accurate estimate of the rigid transformation relating the vehicle and infrastructure sensor frames. This transformation is generally not known a priori and must instead be recovered through point cloud registration.

Point cloud registration has been extensively studied in the scan-to-scan setting, in which two point clouds are acquired within a short time interval, either by the same sensor or by co-located sensors, with small relative displacement or large mutual overlap. The V2I registration problem considered in this work differs substantially from this setting. The vehicle and infrastructure sensors observe the scene from viewpoints separated by tens to hundreds of meters. Their overlap varies with the vehicle's position and is often much smaller than in scan-to-scan settings. Their data acquisition must be temporally synchronized, since each agent operates on an independent clock. Moreover, the initial pose estimate available to any registration method aggregates the localization errors of both agents rather than relying on a single, comparatively accurate source. As a result, methods and initialization strategies developed and validated for scan-to-scan registration cannot be assumed to transfer directly to the V2I setting.

Learning-based registration methods have shown strong performance on established benchmarks, but their ability to generalize to scenarios that differ substantially from their training distributions remains uncertain [1], [2]. This limitation is critical for autonomous driving applications, which must operate reliably in previously unseen environments. Feature-based methods, whether classical or learned, face similar constraints because they rely on distinctive local geometric structure, which is often lacking in structured urban settings with large, geometrically ambiguous surfaces. These considerations favor classical, geometry-based registration methods that do not depend on handcrafted or learned features, are computationally efficient, and tolerate imperfect initialization, pointing toward Generalized ICP and its voxelized variant, VGICP.

While VGICP has been shown to perform well in its original scan-to-scan evaluation setting, its behavior in the V2I setting, under the specific initialization conditions and noise characteristics that this scenario entails, has not been characterized. This work addresses that gap. Specifically, this thesis makes the following contributions:

- We evaluate VGICP on the V2X-Real dataset in both the scan-to-scan and V2I settings, quantifying how much registration accuracy degrades in the latter and identifying the role of scene overlap in this degradation.
- We introduce a controlled noise model that perturbs the initial pose estimate along translational and rotational degrees of freedom independently, reflecting the error characteristics of real GPS/IMU-based localization systems, and use it to characterize the range of initialization error VGICP can tolerate before registration fails.
- We propose a sequential initialization strategy, in which each frame's initial estimate is derived from the previous frame's registered transformation rather than recomputed

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from noisy sensor readings, and evaluate its effect on registration accuracy across sequences of frames.
- We evaluate the effect of VGICP's parameters, voxel resolution and correspondence distance, on registration accuracy and Runtime, characterizing the trade-off between them.
- We validate our findings on real-world sensor data collected by Beamagine, for which no ground-truth poses are available, demonstrating the practical applicability of our approach beyond the V2X-Real dataset.

The remainder of this document is organized as follows. Section II reviews related work on point cloud registration and V2X datasets. Section III formalizes the V2I registration problem, defines the notation used throughout our report, and describes the registration method, initialization strategies, and noise model. Section IV details the experimental design used to evaluate these components. Section V presents and discusses the corresponding results. Finally, Section VI summarizes our findings and outlines directions for future work.

II. State of the Art

A. Point Cloud Registration

Point cloud registration seeks a spatial transformation that best aligns two partially overlapping point clouds. This methods can be broadly grouped into classical geometry-based and learning-based approaches.

a) Classical methods: Classical registration methods can be further split into 4 broad families. The first family [3]–[7] are the algorithms that follow the ICP [8] algorithm, which finds the nearest-neighbor correspondences between the source and target, and computes the optimal rigid transformation that minimises the sum of squared distances over those correspondences. For example, KISS-ICP [6], which relies on point-to-point ICP combined with adaptive thresholding for correspondence matching; or GICP [4], which enhances robustness to point cloud noise by introducing covariance matrices, are part of this family.

A second family [9]–[14] consists of probabilistic methods mostly based on the Normal Distributions Transform [15] (NDT), which discretizes the target point cloud into a voxel grid and models the local point distribution within each voxel as a Gaussian, achieving registration by maximizing the likelihood of the source points under this piecewise-smooth probabilistic representation. There are also probabilistic methods not based on NDT, such as those derived from Coherent Point Drift [16]. For example, D2D-NDT [9], which uses distribution-to-distribution matching instead of point-to-distribution; NDT-ICP [11], which uses NDT for coarse registration and ICP for fine refinement; and Skewed CPD [14], which replaces the isotropic Gaussian from CPD [16] with an asymmetric distribution constructed from local surface normals and curvature radii, are examples of this family.

The third family [17]–[20] are the feature-based methods. These usually follow a two-stage workflow: determining correspondences and estimating the transformation. The correspondence stage typically uses a four-step pipeline: keypoint detection, feature description, descriptor matching, and outlier rejection. In this family, methods are mainly distinguished by their hand-crafted features. For example, FPFH [17], 3DSC [20] and TOLDI [19] are 3D features used in this family.

Finally, the fourth family [21]–[25] are the graph-based methods, which filter correspondences using geometric consistency checks, construct a compatibility graph, and identify inliers through maximum clique detection. For example, TEASER [21], which length invariance under transformation to verify correspondences; CLIPPER+ [22], which is a fast maximal clique algorithm for unweighted graphs enabling robust global registration suitable for real-time applications; and PointTruss [25], which introduces the k-truss from graph theory into point cloud registration, leveraging triangle support as a constraint for inlier selection, belong to this family.

b) Learning-based methods: Learning-based methods can be further split into 3 broad families. The first family [1], [26]–[28] are the feature-based methods, which are the direct learning-based counterpart of the classical feature-based methods. This methods use a deep learning architecture to learn a mapping from point neighborhoods to compact and discriminative feature vectors, and leaves the rest of the classical pipeline untouched. For example, BUFFER-X [1], which aims to improve the generalization and performance of its predecessor BUFFER [26]; COLORPCR++ [27], which adds color information to the extracted features; and UGP [2], which states that cross-attention is harmful for generalization and gets rid of it, belong to this family.

The second family [29]–[34] are the end-to-end methods, which learn how to directly output the rigid transformation from the two raw point clouds, without ever producing a correspondence set. For example, PointNetLK [33], which extends PointNet [35] by incorporating a differentiable Lucas-Kanade [36] layer to iteratively estimate the transformation; and PCRNet [32], which employs a Siamese network architecture to learn global point cloud features and regress the transformation parameters directly via a fully connected regression network, are part of this family.

Finally, the third family [37]–[40] are the generative methods, which cast registration as a generative modeling problem over the SE(3) transformation space. For example, PCRDifussion [38], which proposes a framework that formulates point cloud registration as a denoising diffusion process from noisy transformation to object transformation; and C-GenReg [39], which leverages the complementary strengths of world-scale generative priors and registration-oriented Vision Foundation Models, belong to this family.

In an autonomous driving scenario, we need a method able to generalize to any unseen scenario. Learning-based methods, across all three families discussed above, rely on patterns learned from their training distribution, and their performance on scenarios substantially different from that distribution is not guaranteed; we therefore restrict our choice to classical, geometry-based methods. Among these, we further avoid feature-based and graph-based methods, since both depend on distinctive local geometric structure to compute or match features and correspondences, which is often absent in structured urban environments containing large, geometrically ambiguous surfaces such as plain walls or building facades. Probabilistic

(NDT-based) and ICP-based methods, by contrast, operate directly on the local point or voxel distribution without requiring distinctive geometric features, making them more suitable for such scenarios. Within this remaining group, the method also needs to be fast and tolerant of non-optimal initial transformation guesses, given the sensor-composed localization error present in our vehicle-to-infrastructure registration setting. We choose to work with Generalized ICP (GICP) [4] for its robustness, obtained through a probabilistic point-to-plane formulation that models local surface structure instead of relying on raw point-to-point distances alone. However, standard GICP requires an expensive nearest-neighbor search to establish point correspondences at every iteration, which limits its speed and scalability to large point clouds. We therefore adopt Voxelized GICP (VGICP) [7], which improves on GICP in both speed and robustness through a voxelized approach: aggregating points into voxels avoids the expensive nearest-neighbor search, makes the method parallelizable, and increases robustness to sparser point clouds, since a voxel's covariance can be reliably estimated even when few points from a wall or ground plane fall inside it.

B. Datasets

The development of cooperative perception has been driven by a series of datasets of increasing realism and scale. Early benchmarks such as OPV2V [41] and V2X-Sim [42] were collected entirely in simulation, enabling large-scale data generation but introducing a well-documented sim-to-real gap that limits generalization to real-world deployments [43]. To address this problem, real-world Vehicle-to-Infrastructure (V2I) datasets have been created, such as DAIR-V2X [44] and V2X-Real [45]. However, existing datasets focus primarily on ordinary traffic scenarios, constraining the evaluation of different registration methods to routine situations. CATS-V2V [46] mitigates this limitation by providing a real-world dataset for V2V cooperative perception under complex adverse traffic scenarios.

Because our work focuses on performing registration between vehicles and infrastructure (V2I) in urban scenarios, we use the V2X-Real dataset. While more recent real-world datasets such as CATS-V2V exist, they target vehicle-to-vehicle (V2V) collaboration under adverse traffic scenarios rather than the V2I setting considered in this work, and are therefore not directly applicable here. Among V2I datasets, V2X-Real is substantially more recent than DAIR-V2X (2024, vs. DAIR-V2X's 2022) and is explicitly positioned by its authors as extending beyond DAIR-V2X's single-mode V2I collaboration to a broader multi-agent V2X setting. It also explicitly addresses multi-agent temporal synchronization by GPS-locking each agent's system clock and LiDAR phase, whereas DAIR-V2X's benchmark introduces a dedicated late-fusion mechanism (Time Compensation Late Fusion) specifically to compensate for unresolved temporal asynchrony between vehicle and infrastructure sensors.

III. Method

A. Problem Formulation

Point cloud registration is the task of recovering the rigid transformation that aligns two partially overlapping sets of 3D points. In the vehicle-to-infrastructure (V2I) setting considered in this work, the two point clouds originate from physically distinct sensors observing a shared scene from substantially different viewpoints. This distinguishes the problem from the scan-to-scan formulation for which most registration methods have been developed and benchmarked, and in which the two clouds are acquired at closely spaced instants, either by the same sensor or by different sensors mounted close to each other. This distinction is not merely nominal: the two settings differ substantially along several axes such as:

- **Viewpoint separation**: usually below 2 meters in scan-to-scan, but up to hundreds of meters in the V2I scenario.
- **Overlap**: nearly constant and typically above 90% for consecutive frames in scan-to-scan, but variable and dependent on the vehicle's position relative to the infrastructure LiDAR in the V2I scenario.
- **Temporal synchronization**: not required in scan-to-scan, since a single or master clock is used, but required in the V2I scenario, since independent clocks on each device must be synchronized to register temporally nearby frames.
- **Pose provenance**: in the V2I scenario, the initial guess combines the localization errors of both sensors, unlike in scan-to-scan, where there is a single sensor or the relative position between sensors is known and fixed.

These differences make the V2I problem substantially different from the scan-to-scan setting.

Finally, we state assumptions made throughout all of this work:

- **Rigidity**: The relationship between the two clouds is a single rigid transformation.
- **Metric consistency**: Both clouds are expressed in meters, with the same axis convention, and there is no scale ambiguity between them.
- **Maximum car velocity**: The maximum car speed is assumed to be 50km/h, as this is usually the maximum driving speed allowed within a city.

B. Notation and Reference Frames

First, we define the notation used for transformations between coordinate frames. A transformation or pose $T_{\mathcal{AB}}$ maps coordinates expressed in frame $\mathcal{A}$ to coordinates expressed in frame $\mathcal{B}$, such that for any point p in homogeneous coordinates

$$p_{\mathcal{B}} = T_{\mathcal{AB}}\, p_{\mathcal{A}}, \quad T_{\mathcal{AB}} = \begin{bmatrix} R_{\mathcal{AB}} & t_{\mathcal{AB}} \\ 0^{\top} & 1 \end{bmatrix} \in SE(3)$$

where $p_{\mathcal{A}}$ and $p_{\mathcal{B}}$ denote the point p expressed in frames $\mathcal{A}$ and $\mathcal{B}$, respectively, $R_{\mathcal{AB}} \in SO(3)$ is the rotation component, and $t_{\mathcal{AB}} \in \mathbb{R}^3$ is the translation component.

The rotation $R_{\mathcal{AB}}$ is parameterized by yaw (ψ), pitch (θ), and roll (ϕ), following the intrinsic Z-Y-X convention:

$$R_{\mathcal{AB}} = R_z(\psi)\, R_y(\theta)\, R_x(\phi)$$

where the elementary rotation matrices are

$$R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix},$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}.$$

The translation $t_{\mathcal{AB}}$ is given directly by its Cartesian components,

$$t_{\mathcal{AB}} = \begin{bmatrix} x & y & z \end{bmatrix}^\top \in \mathbb{R}^3.$$

Together, $R_{\mathcal{AB}}$ and $t_{\mathcal{AB}}$ fully determine the pose $T_{\mathcal{AB}}$ through the six parameters $(x, y, z, \psi, \theta, \phi)$, which we use throughout the remainder of this section.

We work with three different reference frames:

- **World frame** ($\mathcal{W}$): A fixed global reference in which the poses of both sensors are expressed. With the V2X-Real dataset, this frame is defined as a right-handed frame with its origin fixed at an arbitrary point (not specified) on the global map formed by the different scenarios, and it is the frame in which ground-truth poses are provided. With the Beamagine data, the frame is defined as a left-handed frame with its origin GPS coordinate zero, since that is what gives us the initial position.
- **Infrastructure frame** ($\mathcal{I}$): A reference attached to the infrastructure LiDAR, with its origin at the sensor optical center and axes that follow the manufacturer convention. Because this sensor is static, its pose $T_{\mathcal{IW}}$ is constant and considered known at any given time.
- **Vehicle frame** ($\mathcal{V}$): A reference attached to the vehicle LiDAR and having its origin at the sensor rather than at the vehicle body reference point. The vehicle sensor pose $T_{\mathcal{VW}}(t)$ varies with time and can be obtained from multiple sources such as Global Navigation Satellite System (GNSS) and Inertial Navigation System (INS) estimates, or the ground truth from the dataset used.

We use $T_{\mathcal{VI}}$ as the target transformation to estimate throughout the rest of the document, for two reasons. First, the infrastructure sensor is static, so $\mathcal{I}$ provides a common reference frame that remains stable across the whole sequence, whereas $\mathcal{V}$ moves between frames. Second, if we want to register multiple vehicles with the same infrastructure, expressing each registration in $\mathcal{I}$ lets us obtain a global registration by registering each vehicle with the infrastructure, without additionally registering every pair of vehicles, since any vehicle-to-vehicle transformation can then be recovered as

$$T_{\mathcal{V}_j\mathcal{V}_i} = T_{\mathcal{V}_i\mathcal{I}}^{-1} T_{\mathcal{V}_j\mathcal{I}},$$

where $\mathcal{V}_i$ and $\mathcal{V}_j$ are the frames of vehicles i and j, respectively.

The remaining notation used throughout the document, including symbols not yet introduced (e.g., T_{err}, $\widehat{T}_{\mathcal{AB}}$), is summarized in Table I.

TABLE I: Notation used throughout the document

Symbol	Meaning
$\mathcal{W}, \mathcal{V}, \mathcal{I}$	World, vehicle and infrastructure frames
$\mathcal{P}_{\mathcal{A}}^{\mathcal{I}}$	Infrastructure point cloud expressed in frame $\mathcal{A}$
$\mathcal{P}_{\mathcal{A}}^{\mathcal{V}}$	Vehicle point cloud expressed in frame $\mathcal{A}$
$T_{\mathcal{AB}} \in SE(3)$	Transformation from frame $\mathcal{A}$ to $\mathcal{B}$
$R_{\mathcal{VI}}, t_{\mathcal{VI}}$	Rotational and translational components of $T_{\mathcal{VI}}$
$T_{\mathcal{AB}}^{*}$	Ground-truth transformation
$\widetilde{T}_{\mathcal{AB}}$	Transformation estimated from sensors
$T'_{\mathcal{AB}}$	Perturbed transformation
$\widehat{T}_{\mathcal{AB}}$	Transformation estimated from VGICP
T_{init}	Initial estimate of $T_{\mathcal{VI}}$
$\widetilde{T}_{\text{init}}$	Perturbed initial estimate of $T_{\mathcal{VI}}$
ΔT	Perturbation of the initial state
$T_{\text{err}} = \hat{T}_{\mathcal{VI}}(T_{\mathcal{VI}}^{*})^{-1}$	Error transformation

C. Registration Method and Initialization

We perform pose registration using Voxelized Generalized ICP (VGICP) [7], as implemented in `small_gicp` [47]. VGICP extends Generalized ICP (GICP) [4] by aggregating point-to-point covariances within a voxel grid rather than computing them per-point, reducing computational cost while retaining the robustness of GICP's probabilistic point-to-plane formulation.

Before registration, we downsample both point clouds using `downsampling_resolution`, fixed to 0.2 m throughout this work, to reduce the number of points, make the registration problem computationally tractable and ensure the data fits in memory.

VGICP's behavior is further governed by a small set of parameters. The `voxel_resolution` controls the granularity of the voxel grid used to aggregate point covariances for correspondence search, while `max_correspondence_distance` sets the maximum distance at which a source point may be matched to a target point or voxel during optimization. Unlike `downsampling_resolution`, these two parameters are varied across our experiments, and their specific values are reported alongside each corresponding experiment.

Beyond these, VGICP exposes additional parameters that control thread parallelism and optimization convergence: maximum number of iterations (`max_iterations`), rotation convergence threshold (`rotation_epsilon`), translation convergence threshold (`translation_epsilon`), and number of threads (`num_threads`). Throughout this work, we use the default values for these parameters, except for `num_threads`, which we set to 4 instead of its default value of 1:

- `max_iterations` $= 20$
- `rotation_epsilon` $= 0.1 \cdot \pi/180 \approx 1.75 \times 10^{-3}$ rad ($0.1°$)
- `translation_epsilon` $= 10^{-3}$ m
- `num_threads` $= 4$ (default: 1)

Because VGICP performs local, iterative optimization, it requires an initial estimate of $T_{\mathcal{VI}}$ to converge to the correct solution. To this end, we define two kinds of initial estimates when working with the V2X-Real dataset:

$$T_{\text{init}} = T^{*-1}_{\mathcal{IW}}\, T^{*}_{\mathcal{VW}}, \qquad \widetilde{T}_{\text{init}} = T^{*-1}_{\mathcal{IW}}\, T'_{\mathcal{VW}}.$$

T_{init} corresponds directly to $T^{*}_{\mathcal{VI}}$, since it is extracted from ground truth. However, as we show later in this document, we examine how different perturbations of this initial estimate, added to mimic the noise that different sensors would introduce in this scenario, affect the final estimate $\widehat{T}_{\mathcal{VI}}$ produced by the method. To this end, we define a second initial estimate, $\widetilde{T}_{\text{init}}$, built from a perturbed vehicle pose $T'_{\mathcal{VW}}$ rather than the ground-truth pose $T^{*}_{\mathcal{VW}}$; the construction of $T'_{\mathcal{VW}}$ is detailed in Section III-E. Note that only the vehicle pose is perturbed, since the infrastructure is treated as a fixed, surveyed anchor; the infrastructure pose $T^{*}_{\mathcal{IW}}$ is therefore kept exact in both T_{init} and $\widetilde{T}_{\text{init}}$.

For the Beamagine data, no ground truth is available, so all poses are computed directly from sensor information. Since these sensor-derived poses already include their own noise, we reuse the notation $\widetilde{T}_{\text{init}}$ for this dataset's initial estimate as well, although here it is computed directly from the noisy sensor poses rather than by perturbing a ground-truth-derived T_{init}:

$$\widetilde{T}_{\text{init}} = \widetilde{T}^{-1}_{\mathcal{IW}}\, \widetilde{T}_{\mathcal{VW}}.$$

When registering a sequence of frames rather than a single pair, we consider two strategies for constructing the initial estimate at each frame k. Under *per-frame initialization*, the initial estimate is constructed independently at every frame, following the definitions of T_{init} and $\widetilde{T}_{\text{init}}$ given above. Under *sequential initialization*, the transformation estimated by VGICP at the previous frame is reused as the initial estimate for the current frame,

$$T^{(k)}_{\text{init}} = \begin{cases} T_{\text{init}} \text{ (or } \widetilde{T}_{\text{init}}), & k = 1, \\ \widehat{T}_{\mathcal{V}(k-1)\mathcal{I}}, & k > 1, \end{cases}$$

where $\widehat{T}_{\mathcal{V}(k-1)\mathcal{I}}$ denotes the transformation estimated by VGICP at frame $k-1$. The first frame of the sequence falls back to per-frame initialization, since no previous estimate is yet available. Sequential initialization is motivated by two considerations. First, the maximum vehicle velocity assumption stated in Section III-A implies that, given the sensor's frame rate, frame-to-frame displacement is bounded, so the previous frame's estimated transformation may remain a reasonable approximation of the current one. Second, propagating a single initial estimate across the sequence means that only one, possibly noisy, sensor reading is required to initialize registration, rather than relying on a fresh, independently noisy reading at every frame. We evaluate this hypothesis in Section IV-C.

D. Dataset

We evaluate our method on the V2X-Real dataset [45], a large-scale cooperative perception dataset that provides synchronized LiDAR data from connected vehicles and roadside infrastructure. Full details on sensor configurations, agent types, and scenario composition are given in the original publication. Each frame provides ground-truth poses $T^{*}_{\mathcal{VW}}$ and $T^{*}_{\mathcal{IW}}$ for the vehicle and infrastructure LiDAR frames relative to the shared map frame $\mathcal{W}$.

Since VGICP requires no training, the standard train/validation/test protocol serves a different purpose in this work than in learning-based methods. For both the initial VGICP evaluation and noise sensitivity analysis (Sections IV-B and IV-C), we perform no parameter selection or optimization, so we combine all three splits to obtain a larger and more statistically robust sample of scenes for characterizing registration behavior across noise levels. When comparing method variants (Section IV-D), we revert to the standard protocol: configurations are selected using the training and validation splits, and final results are reported on the test split only.

To characterize how spatial overlap between the vehicle and infrastructure point clouds affects registration, we additionally compute an *overlap ratio* for each frame. Given the ground-truth transformation $T^{*}_{\mathcal{VI}}$, we transform the vehicle point cloud into the infrastructure frame and measure the proportion of points that have a corresponding point within $1\,\text{m}$ in the infrastructure point cloud. Higher overlap ratios correspond to more spatially well-defined registration problems, since a larger fraction of points have a valid correspondence in the other cloud. Throughout our experiments, we report results both over the full dataset and restricted to frames exceeding overlap thresholds of 30%, 50%, and 80%, to assess how registration accuracy improves as scene overlap increases.

E. Noise Model

To characterize the sensitivity of the registration to initialization error, we deliberately perturb the initial estimate with a controlled transformation before passing to VGICP. This is necessary because the ground-truth poses provided by the dataset are already highly accurate and therefore are not representative of the raw, unrefined readings a real positioning sensor would produce in real time. Perturbing a known initial estimate, rather than relying on the native error of the positioning system, makes the analysis a controlled experiment: the magnitude and affected degrees of freedom are both known exactly, so the resulting degradation can be attributed directly to the perturbed values, allowing us to characterize how the noise introduced by different sensors in this scenario could affect the final registration.

Concretely, given the ground-truth vehicle pose parameters $(x, y, z, \psi, \theta, \phi)$ corresponding to $T^{*}_{\mathcal{VW}}$, we construct the perturbed pose $T'_{\mathcal{VW}}$ by perturbing each parameter independently,

$$x' = x + \Delta x, \quad y' = y + \Delta y, \quad z' = z + \Delta z,$$
$$\psi' = \psi + \Delta\psi, \quad \theta' = \theta + \Delta\theta, \quad \phi' = \phi + \Delta\phi$$

and then reconstructing the pose from these perturbed parameters as

$$T'_{\mathcal{VW}} = \begin{bmatrix} R_z(\psi')\, R_y(\theta')\, R_x(\phi') & [x',\, y',\, z']^{\top} \\ 0^{\top} & 1 \end{bmatrix}$$

The perturbation parameters $\Delta x, \Delta y, \Delta z, \Delta\psi, \Delta\theta, \Delta\phi$ used to construct $T'_{\mathcal{VW}}$ are sampled independently from distributions chosen to reflect the physical characteristics of each channel. Translational noise is modeled with a zero-mean Gaussian distribution,

$$\Delta x, \Delta y \sim \mathcal{N}(0, \sigma_{xy}^2), \qquad \Delta z \sim \mathcal{N}(0, \sigma_z^2),$$

where x and y share a common standard deviation σ_{xy}, while z is assigned a separate standard deviation σ_z, reflecting the typical difference in noise magnitude between horizontal and vertical positioning error. Both standard deviations are expressed in meters, consistent with the units of the translation components of $T_{\mathcal{AB}}$.

By contrast, rotational noise is sampled from a von Mises distribution rather than a Gaussian, to account for the circular nature of angular quantities,

$$\Delta\psi \sim \mathrm{vM}(0, \kappa_\psi), \quad \Delta\theta \sim \mathrm{vM}(0, \kappa_\theta), \quad \Delta\phi \sim \mathrm{vM}(0, \kappa_\phi),$$

with each angle assigned its own concentration parameter κ, reflecting the differing noise characteristics of yaw, pitch, and roll. Here, κ plays a role analogous to the inverse variance of a Gaussian: larger values of κ correspond to angular perturbations more tightly concentrated around zero. This choice of distributions, including the convention of reporting angular spread in degrees rather than directly in terms of κ, follows the noise model proposed by [48]. The specific values of σ_{xy}, σ_z, κ_ψ, κ_θ, and κ_ϕ used in our experiments are detailed and justified in Section IV.

F. Evaluation Metrics

We evaluate registration performance using four complementary metrics: Relative Translation Error (RTE), Relative Rotation Error (RRE), Success Rate, and Runtime. RTE and RRE follow the convention commonly adopted in recent point cloud registration literature [1], [27], [28], and measure registration accuracy. Building on these, we report the Success Rate to summarize registration reliability as a single interpretable value, since RTE and RRE alone do not directly indicate whether a given registration is usable in practice. Finally, we report Runtime to characterize the computational cost of the method, which is a critical factor for applications such as cooperative perception that require near real-time pose alignment between agents.

Relative Translation Error measures the Euclidean distance between the estimated and ground-truth translation,

$$\mathrm{RTE} = \left\| \mathbf{t} - \hat{\mathbf{t}} \right\|_2,$$

where $\mathbf{t}$ and $\hat{\mathbf{t}}$ denote the ground-truth and estimated translation vectors, respectively. Since for our datasets the point clouds are in meters, the units for RTE are also meters.

Relative Rotation Error measures the geodesic distance, in degrees, between the estimated and ground-truth rotation matrices,

$$\mathrm{RRE} = \left| \arccos\left(\frac{\mathrm{tr}(R^\top \hat{R}) - 1}{2} \right) \right|,$$

where R and $\hat{R}$ denote the ground-truth and estimated rotation matrices, respectively.

Building on these two metrics, we additionally report the Success Rate, defined as the proportion of registrations satisfying a joint error threshold,

$$\mathrm{RTE} < \tau_t \text{ and } \mathrm{RRE} < \tau_r,$$

For the RTE threshold, we set $\tau_t = 0.2\,\mathrm{m}$. This follows from the most demanding scenarios in autonomous driving described in [49], where positioning uncertainty must remain below $0.1\,\mathrm{m}$. Assuming this uncertainty independently for each of the two LiDAR sensors involved in registration (vehicle and infrastructure), we set τ_t to twice this value, $0.2\,\mathrm{m}$, beyond which the resulting error is more likely attributable to registration failure than to LiDAR positioning noise alone.

For the RRE threshold, we set $\tau_r = 0.5^\circ$. At a sensing range of $50\,\mathrm{m}$, a rotational error of 0.5° induces a lateral displacement of approximately

$$50 \times \tan(0.5^\circ) \approx 0.44 \text{ m},$$

which we consider an acceptable bound: small relative to the scale of the scene at this range, and comparable in magnitude to the RTE threshold above, while still rejecting registrations with practically meaningful angular misalignment.

Finally, we report Runtime as the average wall-clock time required to execute the VGICP registration step, excluding point cloud loading and preprocessing outside the registration call. Runtime is averaged over all $N = 28{,}430$ point cloud pairs across all splits of the dataset, using a fixed configuration of an AMD EPYC 9224 CPU and 4 threads, with the `small_gicp` [47] implementation of VGICP. While Runtime could be further reduced by increasing the thread count, we fix it at 4 threads for this measurement to ensure the reported values are comparable across all evaluated point cloud pairs. Note that Runtime inherently depends on the number of voxels generated during registration, which is influenced by both the `voxel_size` parameter of the method and the density of the input point clouds; the reported value therefore reflects an average across scenes of varying point cloud density, rather than a fixed per-scene cost. The specific `voxel_size` used may vary across experiments and is stated individually where relevant.

IV. Experiments

A. Experimental Setup

The VGICP Evaluation (Section IV-B) and Noise Analysis (Section IV-C) experiments share a common configuration, allowing results to be directly compared. Following the split-usage policy described in Section III-D, both experiments use the combined train, validation, and test splits of V2X-Real, since neither involves parameter selection or optimization. We set `voxel_resolution` and `max_correspondence_distance` to 1.0, the default values of the `small_gicp` implementation. All other VGICP parameters follow the values stated in Section III-C.

Since VGICP's configuration is fixed across the VGICP Evaluation and Noise Analysis experiments, Runtime does not

vary meaningfully between conditions and carries no information about the questions these experiments are designed to answer, namely, how registration accuracy is affected by scenario type and initialization noise. We therefore report only RTE, RRE, and Success Rate for these experiments; Runtime is instead reported in Section IV-D, where it is directly relevant to comparing the speed-accuracy trade-off across different VGICP configurations.

B. VGICP Evaluation

We first evaluate VGICP in its standard use case: scan-to-scan registration between consecutive LiDAR frames from a single vehicle, the scenario for which VGICP was originally designed and evaluated [7]. Given two temporally consecutive vehicle poses $T^*_{\mathcal{V}(t)\mathcal{W}}$ and $T^*_{\mathcal{V}(t+1)\mathcal{W}}$, we register the corresponding point clouds $\mathcal{P}^{\mathcal{V}}_{\mathcal{V}(t)}$ and $\mathcal{P}^{\mathcal{V}}_{\mathcal{V}(t+1)}$, evaluating VGICP both with and without a meaningful initial estimate. In the initialized case, we use the ground-truth relative pose

$$T_{\text{init}}^{\mathcal{V}(t)\mathcal{V}(t+1)} = T^{*-1}_{\mathcal{V}(t)\mathcal{W}}\, T^{*}_{\mathcal{V}(t+1)\mathcal{W}},$$

while in the uninitialized case we instead use the identity matrix, $T_{\text{init}} = I_4$. Comparing these two cases evaluates VGICP's accuracy and convergence behavior in the scan-to-scan setting, serving as a baseline check that VGICP performs as expected before proceeding to the V2I setting.

We then evaluate VGICP in the V2I setting central to this work, registering point cloud pairs $\mathcal{P}^{\mathcal{V}}_{\mathcal{V}}$ and $\mathcal{P}^{\mathcal{I}}_{\mathcal{I}}$ from the vehicle and infrastructure sensors, respectively. As in the scan-to-scan case, we evaluate both with and without initialization: the initialized case uses T_{init} as defined in Section III-C, while the uninitialized case again uses $T_{\text{init}} = I_4$. This comparison serves two purposes: first, contrasting the initialized and uninitialized V2I results with their scan-to-scan counterparts quantifies how substantially the V2I setting differs from the scan-to-scan formulation for which VGICP and most registration methods are designed, as discussed in Section III-A; second, it tests whether, given a sufficiently accurate initialization, VGICP can still achieve results of practical use in the V2I setting.

C. Noise Analysis

Using the noise model defined in Section III-E, we evaluate VGICP's registration accuracy under controlled perturbations of the initial estimate. We consider several perturbation types independently: planar displacement, height displacement, yaw, roll, and pitch, as well as the combination of planar displacement and yaw, which we later argue is the pairing most representative of the noise expected in this scenario. Rather than a full sweep, we evaluate two noise magnitudes per perturbation type, a lower and an upper bound, to characterize VGICP's behavior at both conservative and more demanding noise levels, measuring RTE, RRE, and Success Rate in each case.

For planar and height displacement, we adopt the positional noise magnitudes proposed by [48]. The same work also proposes rotational noise magnitudes of 12.5° and 25° for each rotational axis; we evaluate these once for completeness, to observe how such large orientation errors affect registration, but consider them substantially larger than what is realistic for this scenario, corresponding to orientation errors an order of magnitude beyond what typical IMU- and GNSS-based localization systems exhibit in practice. We therefore introduce our own, more realistic rotational noise bounds instead, derived as described below, and use these throughout the remainder of this work.

For roll and pitch, we were unable to find a consistent, sensor-independent characterization of typical error magnitudes in the literature, as these largely depend on the specific IMU used. We therefore set a lower bound of 0.1° and an upper bound of 1°, a range within which most reported IMU orientation errors appear to fall or remain below. For yaw, whose error is additionally affected by GPS-based heading estimation and exhibits substantially higher variance, we set a wider range of 1° to 5°.

For planar displacement, the bounds are further supported by the maximum vehicle velocity assumption in Section III-A. At the assumed maximum velocity of 50 km/h and the dataset's LiDAR frame rate of 10 Hz, the maximum frame-to-frame displacement is

$$d_{\max} = \frac{50 \text{ km/h}}{10 \text{ Hz}} \approx 1.39 \text{ m},$$

which falls within our chosen lower and upper bounds of 1 m and 2 m. The height displacement bound is set equal to the planar displacement bound, in the absence of a scenario-specific justification for treating it differently.

We additionally test whether sequential initialization (Section III-C) is viable in this scenario. Since sequential initialization relies on the previous frame's estimated transformation remaining within VGICP's convergence basin, the perturbation evaluation described above first establishes whether the displacement bound implied by the maximum vehicle velocity assumption falls within a range VGICP can reliably handle. We then directly evaluate sequential initialization across sequences of frames, measuring RTE and RRE per frame to assess whether registration error compounds, remains stable, or is reduced as the sequence progresses.

D. Model Variations

Unlike the VGICP Evaluation and Noise Analysis experiments (Section IV-A), the experiments in this subsection involve selecting between multiple configurations. Following standard practice and consistent with the split-usage policy described in Section III-D, we select parameters and configurations using the training and validation splits, and report all final results in this subsection on the test split only.

To ensure the selected configuration is representative of the noisy initialization conditions under which this method is intended to operate, rather than the idealized ground-truth initialization used in VGICP evaluation, all experiments in this subsection are conducted under the combined planar-and-yaw noise regime established in Section V-B, evaluated separately at its lower and upper bounds.

Beyond the default parameter values used in the previous experiments (Section IV-A), there is limited prior work studying VGICP specifically in the V2I setting, and consequently no established guidance on which `voxel_resolution` value is best suited to it. To address this, we sweep `voxel_resolution` from 0.2 m to 2.0 m in steps of 0.2 m, evaluating Success Rate at each value under both noise bounds, with `max_correspondence_distance` fixed at its default value (1.0). We initially intended to additionally sweep `max_correspondence_distance` jointly with `voxel_resolution`; however, we found that the installed version of `small_gicp` (the latest one at the moment of writing) contains a defect preventing this parameter from being set to any value other than its default, regardless of the value specified. As correcting this and repeating the search was not feasible within the time available for this work, `max_correspondence_distance` is fixed at 1.0 throughout this section, and its effect on V2I registration accuracy is left as an open question. We select the `voxel_resolution` that maximizes Success Rate, since this directly reflects the proportion of registrations achieving errors below the thresholds defined in Section III-F, rather than only the central tendency of the error distributions. The selected value is used as the fixed `voxel_resolution` for all subsequent variations evaluated in this subsection.

We additionally report Runtime (Section III-F) for each configuration evaluated in this subsection. Unlike VGICP Evaluation and Noise Analysis, where VGICP's configuration is held fixed and Runtime therefore carries no information about the questions under study, the variations evaluated here directly alter the size and structure of the point clouds passed to VGICP (e.g., through BEV projection or ground-plane removal) and the parameters governing its internal voxelization, both of which are expected to affect computational cost. Reporting Runtime alongside accuracy therefore allows us to characterize the trade-off between registration accuracy and computational cost across configurations, rather than treating accuracy in isolation.

We evaluate several variations of the registration pipeline aimed at improving accuracy in the V2I setting, defined along two independent factors: the *point cloud* passed to VGICP (`Full`, the original unprojected point clouds; or `BEV`, obtained by projecting all points onto a common height) and whether the *ground plane* is removed prior to registration. First, we evaluate registration performed entirely in BEV, registering directly in this reduced representation. This tests whether reducing the effective dimensionality of the problem, at the cost of discarding vertical structure and height information entirely, improves registration accuracy on its own. Second, rather than using the BEV result directly, we use it only as a coarse initial estimate for subsequent registration performed on the original, unprojected point clouds (`BEV, refined on full`); this tests whether BEV registration is useful specifically as an initialization step, recovering an approximate planar alignment cheaply before a full 3D registration refines it, including the height component that the BEV-only variation discards.

A potential drawback of the BEV projection is that points belonging to the ground plane become intermixed with points from other structures once projected to a common height, which may lead VGICP to establish incorrect correspondences between nearby but semantically unrelated points. To address this, we repeat both BEV-based variations after first removing the ground plane from each point cloud, so that only points above the ground are projected and the resulting BEV representation is less prone to this intermixing. In total, we evaluate six configurations, obtained by combining each point cloud type (`Full`, `BEV`, `BEV, refined on full`) with whether the ground plane is removed (`No`, `Yes`); these are reported in Tables III and IV using this same naming.

E. Beamagine

We apply the best-performing configuration identified in Section IV-D to real-world data collected in the context of the 6GEWOC project, in collaboration with Beamagine, using LiDAR sensors mounted on the vehicle and infrastructure.

Unlike V2X-Real, this data does not provide ground-truth poses. In principle, an initial estimate could be constructed directly from the positioning outputs of the vehicle and infrastructure sensors, following the same approach used to define T_{init} in Section III-C. However, we found the vehicle's positioning sensor to exhibit errors of up to 20 m, well beyond what VGICP can reasonably be expected to recover from directly (Section V-B). We therefore instead construct a coarse manual initial estimate for the first frame of the sequence, by visually aligning the two point clouds in a 3D viewer, and rely on the sequential initialization strategy proposed in Section III-C to propagate this estimate across all subsequent frames, avoiding any further dependence on the vehicle's unreliable positioning data.

V. Results

A. VGICP Evaluation

All results in this section are presented as boxplots, since the underlying error distributions contain a substantial number of outliers. Outliers are defined using the standard $1.5 \times$ IQR criterion; whiskers and quartiles are computed from the full, unfiltered data, and only the outlier points themselves are omitted from the plotted range. Figure 1 shows the scan-to-scan RTE results with outliers included, illustrating that their presence makes the underlying distribution difficult to read. For this reason, we omit outliers from all boxplots in the remainder of this section; the full distributions are reported in Appendix A, which lists the proportion of outliers for each experiment and overlap threshold, along with the corresponding sample sizes. To complement these distributions, we also report the Success Rate (Section III-F) for each condition, shown as bar plots with one bar per overlap threshold.

We first evaluate VGICP in the scan-to-scan setting using ground-truth initialization. As shown in Figure 2, both RTE and RRE remain low across all overlap thresholds, confirming that VGICP performs as expected in the setting for which it was originally designed. Repeating this evaluation with the identity matrix as initialization produces nearly identical results, which is expected: frame-to-frame displacement is

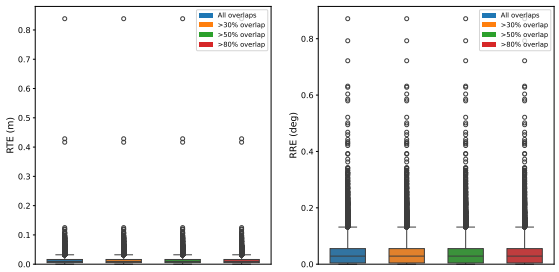


Fig. 1: RTE and RRE for scan-to-scan registration including outliers, shown once to illustrate their impact on plot legibility.

small (Section IV-C) and overlap between consecutive scan-to-scan frames consistently exceeds 80%. This is reflected in the fact that the results are effectively unchanged across all overlap thresholds. This is further confirmed by the Success Rate (Figure 3), which is 99.9% across all overlap thresholds for both initialization conditions. Given how uniformly high these values are, there is little more to discuss for this setting.

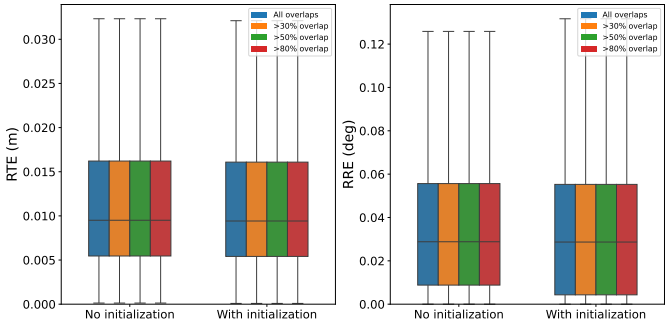


Fig. 2: RTE and RRE for scan-to-scan registration, with ground-truth and identity initialization, across overlap thresholds.

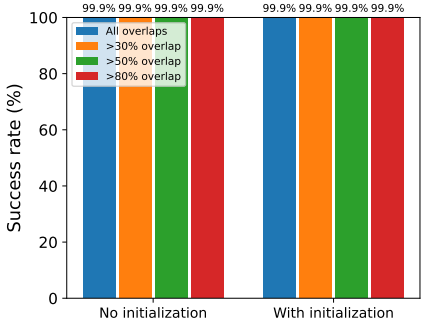


Fig. 3: Success Rate for scan-to-scan registration, with ground-truth and identity initialization, across overlap thresholds.

We next evaluate VGICP in the V2I setting central to this work. With the initial estimate T_{init} derived from ground-truth poses (Section III-C), RTE and RRE (Figure 4) are an order of magnitude higher than in the scan-to-scan setting, despite using this exact initial estimate. We attribute this to the larger sensor-to-sensor distances and the correspondingly lower point density at ranges typical of the V2I setting: since `voxel_resolution` is fixed (Section IV-A), voxels populated by sparser, farther-range points yield less reliable covariance estimates than in the denser scan-to-scan case, degrading correspondence quality. This is consistent with the observed trend across overlap thresholds: as overlap increases, error decreases. The Success Rate (Figure 5) shows a more pronounced version of this trend, with a 23.8 percentage point jump between overlaps above 50% and above 80%, confirming that registration performance improves substantially once scenes are restricted to sufficiently well-overlapping cases.

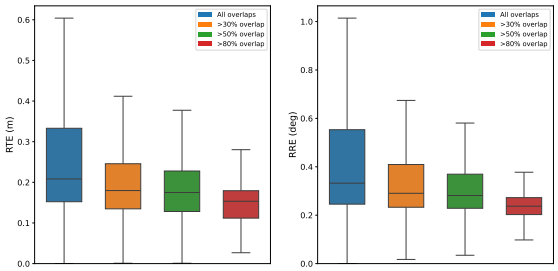


Fig. 4: RTE and RRE for V2I registration with ground-truth initialization, across overlap thresholds.

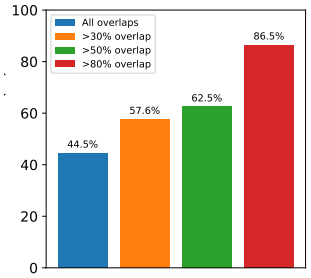


Fig. 5: Success Rate for V2I registration with ground-truth initialization, across overlap thresholds.

Finally, we repeat the V2I evaluation using the identity matrix as initialization (Figure 6). Because errors in this condition are roughly three orders of magnitude larger than in the initialized case, we report it in a separate figure to preserve the legibility of both. The contrast with the initialized case is stark: RTE reaches up to $200\,\text{m}$ and RRE up to $180°$ for the worst cases, and the Success Rate is 0% across all overlap thresholds. We therefore omit a corresponding bar plot, since it would show no variation. This is consistent with the identity transformation being far from the true transformation in this setting, given the substantial difference in viewpoint and position between the vehicle and infrastructure sensors (Section III-A). Under such a poor initial estimate, VGICP's local optimization is prone to converging to a poor local minimum, or in the most extreme cases, failing to establish meaningful correspondences at all. This confirms the concern

motivating this evaluation: as a local optimization method, VGICP is not designed to recover from initial estimates far from the true transformation, and the identity-initialized V2I results here illustrate the practical consequence of violating that requirement.

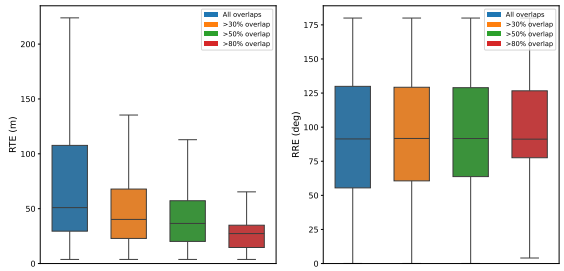


Fig. 6: RTE and RRE for V2I registration with identity initialization, across overlap thresholds.

B. Noise Analysis

We first look at planar noise (perturbing x and y) at the lower and upper bounds of 1 m and 2 m, respectively, with results summarized in Table II alongside all other conditions evaluated in this section.

1) Planar Noise: At the lower bound, translation error increases relative to the no-noise case, as shown in Table II, but VGICP still recovers registrations with errors comparable to those obtained without noise. Rotational error is essentially unaffected, as expected, since only planar coordinates are perturbed. Success Rate drops accordingly, with a maximum reduction of 3.6% for the $> 50\%$ overlap regime. Notably, this reduction grows with the overlap threshold, likely because at lower overlap thresholds the Success Rate is already reduced by limited correspondences alone, leaving comparatively less headroom for the added noise to further degrade it.

At the upper bound of 2 m, translation error increases substantially: maximum errors across most overlap thresholds are more than five times larger than under the lower-bound noise. As Table II shows, the reported values reach translation errors above $2 \times \sigma_{xy}$ for all overlap thresholds except $> 80\%$, and above $1.5 \times \sigma_{xy}$ (3 m) for the $> 80\%$ threshold, where $\sigma_{xy} = 2$ m. This indicates that, beyond typical noise magnitudes, VGICP is no longer simply absorbing the perturbation but converging to incorrect registrations. Rotational error also increases under this noise level, though only modestly, and Success Rate drops substantially further relative to the lower-bound case.

2) Height Noise: We then introduce height noise, at the same 1 m and 2 m bounds. As Table II shows, the resulting errors and the corresponding reduction in Success Rate are considerably smaller than those caused by planar noise, particularly when comparing the two upper bounds.

3) Angular Noise: We next consider angular noise, starting with the magnitudes proposed by [48] ($12.5°$ and $25°$, applied independently to roll, pitch, and yaw). These magnitudes introduce errors large enough to substantially degrade VGICP's ability to converge to low-error registrations, as reflected in the Success Rate values in Table II, confirming our expectation that they are too large to be realistically handled by the method in its current form. Roll is consistently the least affected of the three angles, while pitch and yaw behave similarly at the lower magnitude; at the upper magnitude, yaw shows the most pronounced degradation, particularly at higher overlap thresholds. We attribute yaw's comparatively worse behavior at the upper bound to yaw perturbations displacing points increasingly with distance from the point cloud origin; since these far-range points are already sparse due to their distance from the LiDAR, large yaw perturbations prevent the algorithm from establishing correct correspondences for them, or cause it to establish incorrect ones. These results support our decision to introduce more realistic rotational noise bounds.

We then evaluate the more realistic noise magnitudes proposed in Section III-E. Roll and pitch are considered together, as their results are similar in Table II: for these more realistic noise magnitudes, errors increase only slightly, with a maximum Success Rate reduction of approximately 1.1% for the regime with all overlaps, relative to the no-noise case, even at the upper bound.

For yaw, the lower bound behaves similarly to the upper bound of roll and pitch, with a modest increase in error and a small reduction in Success Rate. The upper bound, however, behaves differently: errors are substantially higher and exhibit a considerably wider interquartile range than any other realistic noise condition examined, for both translation and rotation. As with the paper-proposed yaw noise, we attribute this to yaw perturbations displacing points increasingly with distance from the origin, producing a wide range of outcomes across the sampled yaw values. Despite this wide error range, Success Rate does not decrease as much as the error magnitudes alone would suggest, remaining comparable to the reduction observed under height noise. This indicates that, although errors span a wide range, they remain concentrated toward the lower end of that range for both RTE and RRE.

4) Planar and Yaw Noise: Since planar and yaw noise introduce the largest errors and are also the hardest to constrain, as both originate from GPS-based estimation, we adopt them as the representative noise sources for the remainder of this work and exclude the remaining noise types from further analysis. We evaluate their joint effect by combining the lower bounds of both (1 m and $1°$ for planar and yaw noise, respectively), and separately the upper bounds of both (2 m and $5°$ for planar and yaw noise, respectively). As shown in Table II, at the lower bound errors are slightly larger than under planar noise alone, but the increase is small and Success Rate is not substantially reduced relative to the planar-only case; since yaw's lower bound alone was already close to the no-noise scenario, this suggests planar noise remains the dominant source of error in this regime, with yaw contributing a comparatively smaller effect. At the upper bound, however, planar and yaw noise appear to interact, increasing the errors and substantially reducing Success Rate across all overlap regimes.

5) Sequential Initialization: Finally, we evaluate sequential initialization (Section III-C) under the combined planar-and-

TABLE II: RTE and RRE quartiles (Q_1/median/Q_3, in meters and degrees, respectively) and Success Rate (%, in parentheses) per noise condition and overlap threshold.

Condition	All	$> 30\%$	$> 50\%$	$> 80\%$
Planar, lower (1 m)	0.16/0.22/0.40 – 0.25/0.34/0.59 (41.5%)	0.14/0.19/0.27 – 0.23/0.30/0.43 (54.3%)	0.13/0.18/0.24 – 0.23/0.29/0.38 (58.9%)	0.11/0.16/0.19 – 0.20/0.24/0.28 (83.0%)
Planar, upper (2 m)	0.18/0.35/2.33 – 0.26/0.41/0.77 (29.3%)	0.16/0.25/1.91 – 0.25/0.34/0.60 (38.4%)	0.16/0.23/1.79 – 0.24/0.32/0.52 (41.7%)	0.13/0.18/1.37 – 0.21/0.25/0.35 (60.3%)
Height, lower (1 m)	0.15/0.21/0.35 – 0.25/0.34/0.58 (43.4%)	0.14/0.18/0.25 – 0.23/0.29/0.42 (56.7%)	0.13/0.18/0.23 – 0.23/0.28/0.37 (61.5%)	0.11/0.15/0.18 – 0.20/0.24/0.28 (85.2%)
Height, upper (2 m)	0.17/0.26/0.93 – 0.26/0.38/0.76 (36.0%)	0.15/0.20/0.37 – 0.24/0.32/0.59 (47.1%)	0.14/0.20/0.31 – 0.24/0.31/0.49 (51.0%)	0.12/0.17/0.21 – 0.21/0.25/0.32 (70.4%)
Roll (paper), 12.5°	0.16/0.24/0.74 – 0.26/0.39/1.24 (37.8%)	0.15/0.20/0.32 – 0.24/0.32/0.62 (49.4%)	0.14/0.19/0.28 – 0.24/0.30/0.49 (53.6%)	0.12/0.17/0.20 – 0.21/0.25/0.31 (73.8%)
Roll (paper), 25°	0.19/0.51/5.39 – 0.31/0.89/20.48 (25.7%)	0.17/0.29/4.18 – 0.28/0.59/19.70 (33.7%)	0.17/0.26/3.95 – 0.27/0.47/19.43 (36.6%)	0.14/0.20/3.68 – 0.23/0.31/19.70 (49.5%)
Pitch (paper), 12.5°	0.20/4.61/13.99 – 0.30/5.55/13.20 (25.0%)	0.17/1.83/9.13 – 0.26/3.17/12.23 (32.8%)	0.17/0.74/7.98 – 0.25/1.57/11.76 (35.6%)	0.13/0.18/2.67 – 0.22/0.27/8.84 (57.4%)
Pitch (paper), 25°	1.66/9.98/29.10 – 3.51/14.46/25.68 (14.8%)	0.28/6.59/18.67 – 0.49/13.61/24.90 (19.4%)	0.23/5.69/15.87 – 0.37/13.22/24.40 (21.1%)	0.17/2.23/7.58 – 0.25/10.74/22.14 (34.4%)
Yaw (paper), 12.5°	0.40/6.05/16.02 – 0.63/7.31/13.88 (16.6%)	0.22/4.21/10.62 – 0.34/6.57/13.13 (21.8%)	0.20/3.80/9.33 – 0.31/6.26/12.96 (23.7%)	0.17/1.47/5.12 – 0.24/3.46/12.06 (38.8%)
Yaw (paper), 25°	4.17/12.91/32.22 – 6.16/16.03/28.34 (9.2%)	3.08/9.26/22.15 – 5.29/15.49/28.01 (12.0%)	2.76/8.47/19.81 – 5.00/15.24/27.82 (13.1%)	0.28/5.04/11.46 – 0.44/14.23/27.24 (21.9%)
Roll, 0.1°	0.15/0.21/0.34 – 0.25/0.33/0.55 (44.3%)	0.13/0.18/0.25 – 0.23/0.29/0.41 (57.6%)	0.13/0.17/0.23 – 0.23/0.28/0.37 (62.4%)	0.11/0.15/0.18 – 0.20/0.24/0.27 (86.5%)
Roll, 1°	0.16/0.21/0.37 – 0.25/0.34/0.63 (43.4%)	0.14/0.18/0.25 – 0.23/0.29/0.42 (56.9%)	0.13/0.18/0.23 – 0.23/0.28/0.38 (61.7%)	0.11/0.15/0.18 – 0.20/0.24/0.27 (86.4%)
Pitch, 0.1°	0.15/0.21/0.33 – 0.25/0.33/0.55 (44.3%)	0.13/0.18/0.25 – 0.23/0.29/0.41 (57.6%)	0.13/0.17/0.23 – 0.23/0.28/0.37 (62.5%)	0.11/0.15/0.18 – 0.20/0.24/0.27 (86.5%)
Pitch, 1°	0.15/0.21/0.34 – 0.25/0.33/0.57 (44.0%)	0.13/0.18/0.25 – 0.23/0.29/0.41 (57.6%)	0.13/0.17/0.23 – 0.23/0.28/0.37 (62.4%)	0.11/0.15/0.18 – 0.20/0.24/0.27 (86.5%)
Yaw, 1°	0.16/0.21/0.38 – 0.25/0.34/0.62 (43.2%)	0.14/0.18/0.25 – 0.23/0.29/0.42 (56.5%)	0.13/0.18/0.23 – 0.23/0.28/0.37 (61.3%)	0.11/0.15/0.18 – 0.20/0.24/0.27 (86.3%)
Yaw, 5°	0.18/0.59/5.98 – 0.27/0.70/5.02 (31.2%)	0.16/0.23/3.85 – 0.24/0.37/3.81 (41.0%)	0.16/0.21/3.13 – 0.24/0.33/3.27 (44.5%)	0.12/0.17/0.21 – 0.21/0.25/0.33 (70.1%)
Planar+Yaw, 1 m/1°	0.16/0.23/0.59 – 0.25/0.35/0.67 (40.5%)	0.14/0.19/0.28 – 0.23/0.30/0.45 (53.1%)	0.14/0.18/0.25 – 0.23/0.29/0.40 (57.6%)	0.11/0.16/0.19 – 0.20/0.24/0.28 (82.6%)
Planar+Yaw, 2 m/5°	0.21/2.84/6.82 – 0.31/1.35/5.10 (22.5%)	0.18/1.79/4.81 – 0.27/0.61/4.12 (29.6%)	0.18/1.38/4.43 – 0.26/0.50/3.80 (32.1%)	0.15/0.19/2.65 – 0.22/0.28/1.39 (51.3%)

Each cell reports RTE Q_1/median/Q_3 [m] – RRE Q_1/median/Q_3 [$^\circ$] (Success Rate %).

yaw noise regime. The number of sequences containing at least N frames, as a function of frame index N, is shown in Figure 7. Since sequential initialization is evaluated per frame, we redefine the overlap regimes so that each one corresponds to continuous sequences in which every frame falls within that overlap threshold, ensuring that frames with lower overlap do not affect the evaluation of higher-overlap regimes. We additionally require sequences to contain at least 10 frames to be considered, so that results are statistically meaningful. As Figure 7 shows, the number of sequences decreases sharply with N, meaning that later frame indices are supported by comparatively few sequences; this should be kept in mind when interpreting the results at the far end of the frame range, as discussed below.

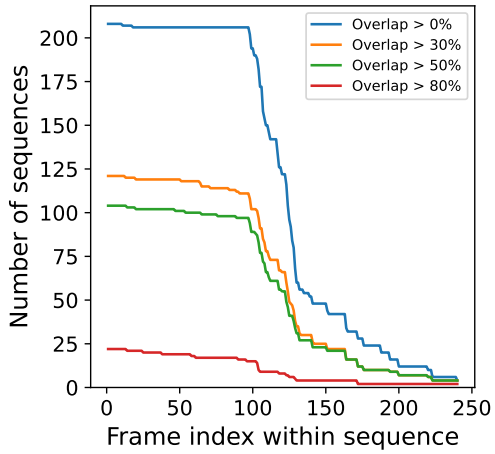


Fig. 7: Number of sequences containing at least N frames, as a function of frame index N, for the combined planar-and-yaw noise regime.

RTE and RRE are reported as the per-frame median, together with a shaded region spanning the first and third quartiles, analogous to a continuous boxplot. We use the median rather than the mean here because, as discussed above, outliers tend to be orders of magnitude larger than typical errors and would make the mean an unreliable representation of typical behavior. Success Rate and overlap ratio, by contrast, are reported as mean values.

For the lower bound of the combined noise, the median error does not decrease substantially across frames in Figure 8. However, for every overlap regime other than the full dataset, the interquartile range narrows as the sequence progresses. This is reflected in the Success Rate trend in Figure 9, which increases across all overlap regimes over the initial frames. This increase is not explained by a corresponding rise in overlap ratio, which remains largely stable over these initial frames, as shown in Figure 12.

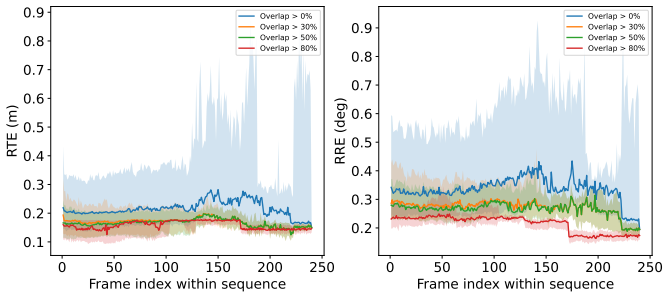


Fig. 8: Per-frame median RTE and RRE, with interquartile range, under sequential initialization at the lower bound of the combined planar-and-yaw noise, across overlap regimes.

For the upper bound, the same trend is substantially more pronounced: error decreases markedly over the initial frames in Figure 10, translating into a correspondingly large increase in Success Rate over the same range, as shown in Figure 11.

Beyond this initial improvement, both Success Rate trends show a subsequent decline starting around frame 50 and continuing to approximately frame 150, before the sparse-sequence effects discussed above begin to dominate. As with the earlier improvement, this decline is not accompanied by any corresponding change in overlap ratio (Figure 12) large enough to explain it for most overlap regimes, with one exception: in the regime including all overlaps, overlap ratio does decrease over this range, while in the remaining, more restrictive overlap regimes it remains stable or shows no significant decrease. This suggests that, for the restricted overlap regimes at least, neither trend is primarily driven by

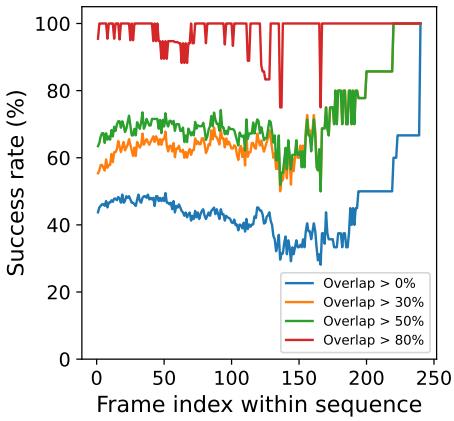


Fig. 9: Mean Success Rate per frame under sequential initialization at the lower bound of the combined planar-and-yaw noise, across overlap regimes.

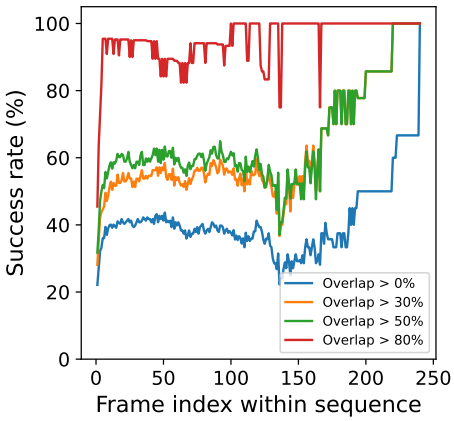


Fig. 11: Mean Success Rate per frame under sequential initialization at the upper bound of the combined planar-and-yaw noise, across overlap regimes.

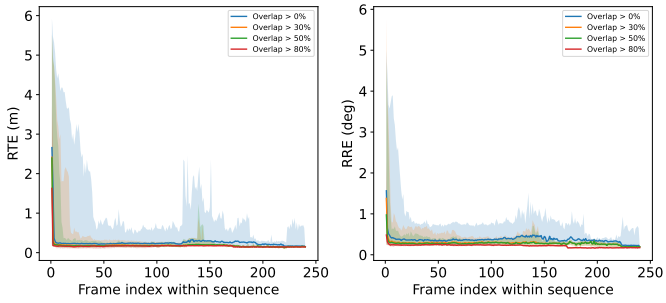


Fig. 10: Per-frame median RTE and RRE, with interquartile range, under sequential initialization at the upper bound of the combined planar-and-yaw noise, across overlap regimes.

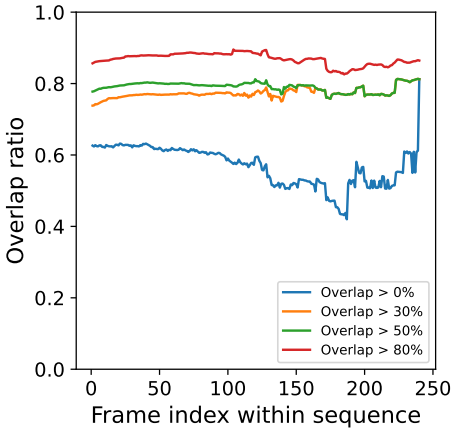


Fig. 12: Mean overlap ratio per frame under sequential initialization, for the combined planar-and-yaw noise regime, at both the lower and upper bounds.

changes in scene geometry, but instead by the behavior of sequential initialization itself; for the regime including all overlaps, however, the decreasing overlap ratio may itself contribute to the observed decline in Success Rate. We attribute the remaining, geometry-independent portion of this decline to an accumulation of drift: because each frame's initial estimate depends on the previous frame's registered transformation, small errors introduced at any point in the sequence can propagate and compound over subsequent frames, gradually degrading registration quality even as overlap remains stable. This is consistent with the same mechanism that produces the narrowing interquartile range at the start of the sequence working in reverse over a longer horizon: whereas early frames benefit from the accumulation of increasingly refined estimates, later frames within this range appear instead to accumulate compounding error. Despite this decline, Success Rate does not drop below its initial, first-frame value at any point within this range, indicating that sequential initialization remains no worse than per-frame initialization even as its benefit diminishes.

6) Sequential Initialization (first order): Motivated by this decline, we explore whether it can be mitigated by using a more informed model of the vehicle's motion between frames, rather than assuming the vehicle remains exactly at its previous position. Specifically, we consider a first-order approximation, in which the vehicle is assumed to move with constant velocity between consecutive frames: rather than reusing the previous frame's registered transformation directly, we extrapolate it using the displacement observed between the two preceding frames, under the assumption that this displacement will approximately repeat. Since the underlying cause of the decline is that each frame's initial estimate falls farther from the true transformation as small errors accumulate, providing an initial estimate that also accounts for the vehicle's expected motion, rather than treating it as stationary between frames, may keep the initial estimate closer to the true transformation for longer, potentially slowing or offsetting this accumulation.

We refer to the strategy evaluated above, which reuses the previous frame's transformation directly, as *zero-order* sequential initialization, and to this constant-velocity extrapolation as *first-order* sequential initialization. Figures 13 and 14 show Success Rate under first-order sequential initialization, for the lower and upper bounds of the combined noise regime, respectively. For the lower bound, the Success Rate does

not differ substantially from that observed under zero-order sequential initialization, suggesting that constant-velocity extrapolation offers little benefit when the underlying drift is already small. For the upper bound, however, first-order sequential initialization appears to partially correct the decline observed previously: the decrease in Success Rate for the regime including all overlaps is less pronounced than under zero-order initialization, and the increasing trend observed over the initial frames for the remaining overlap regimes is stabilized rather than followed by a subsequent decline.

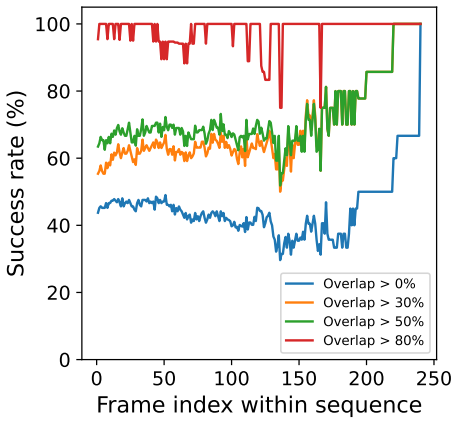


Fig. 13: Mean Success Rate per frame under first-order (constant-velocity) sequential initialization at the *lower bound* of the combined planar-and-yaw noise, across overlap regimes.

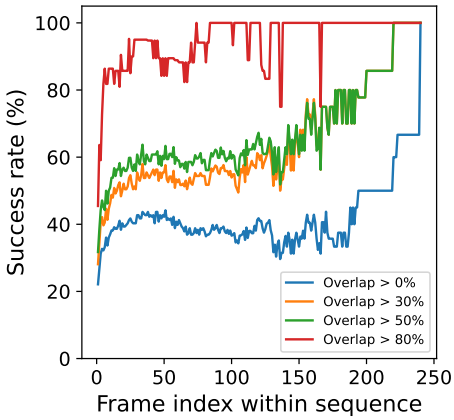


Fig. 14: Mean Success Rate per frame under first-order (constant-velocity) sequential initialization at the *upper bound* of the combined planar-and-yaw noise, across overlap regimes.

While both the lower- and upper-bound Success Rate trends appear to approach 100% for the largest frame indices, Figure 7 shows that very few sequences are long enough to contribute to these final frames. This apparent convergence to a perfect Success Rate should therefore be interpreted with caution, as it may reflect the behavior of a small, potentially unrepresentative subset of sequences rather than a genuine trend.

Taken together, these results show that sequential initialization reduces registration error over the initial frames of a sequence and that a single accurate initial estimate can be propagated across multiple frames without incorporating additional positional data. However, this benefit is not indefinite: registration quality subsequently declines over an intermediate range of frame indices, likely due to accumulating drift, before the small number of surviving long sequences makes later trends unreliable to interpret. This decline can be partially mitigated by extrapolating the vehicle's motion under a constant-velocity assumption rather than treating it as stationary between frames: while this first-order strategy offers little benefit under the lower bound of the combined noise regime, it stabilizes the declining trend under the upper bound, indicating that a more informed motion model can, at least partially, counteract the accumulation of drift responsible for this decline.

C. Model Variations

Figures 15 and 16 show Success Rate as a function of `voxel_resolution` for the lower and upper bounds of the combined planar-and-yaw noise regime, respectively. In both cases, performance plateaus around a `voxel_resolution` of $1\,\text{m}$, beyond which further increases yield no substantial improvement. We hypothesize that this plateau is related to `max_correspondence_distance` being fixed at its default value of $1\,\text{m}$ throughout this sweep (Section IV-D): once `voxel_resolution` approaches this value, points or voxels separated by more than `max_correspondence_distance` cannot be matched regardless of how the voxel grid is further coarsened. This constraint could prevent Success Rate from improving further even if a larger `voxel_resolution` would otherwise be beneficial. Testing this hypothesis directly would require jointly varying both parameters, which, as discussed in Section IV-D, was not possible in this work due to the `small_gicp` defect affecting `max_correspondence_distance`.

For the lower bound, all overlap regimes except $> 80\%$ reach their maximum Success Rate at `voxel_resolution` $= 1.2\,\text{m}$, while the $> 80\%$ regime reaches its maximum at `voxel_resolution` $= 1\,\text{m}$; for the upper bound, all overlap regimes reach their maximum at `voxel_resolution` $= 1.2\,\text{m}$. Since this value maximizes Success Rate across all overlap regimes under the upper bound and across all but one under the lower bound, we select `voxel_resolution` $= 1.2\,\text{m}$ as the configuration used for the remaining variations evaluated in this subsection.

Using this configuration, we evaluate the six pipeline variations described above. Tables III and IV report RTE, RRE, Success Rate, and Runtime for each. Overall, every variation performs worse than the baseline while also increasing Runtime; their comparison, however, still indicates why.

The clearest pattern is that every configuration relying solely on BEV registration, without subsequent refinement on the full point clouds, performs substantially worse than the baseline. This indicates that discarding the height dimension does not simplify the registration problem in this setting, as we had hypothesized, but is instead detrimental. This conclusion is reinforced by the configurations that do refine the BEV

TABLE III: RTE and RRE quartiles (Q_1/median/Q_3, in meters and degrees, respectively) and Success Rate (%, in parentheses) per pipeline configuration and overlap threshold.

Point Cloud	Ground removed	All	$> 30\%$	$> 50\%$	$> 80\%$
Full	No	0.11/0.18/0.36 – 0.21/0.30/0.56 (52.5%)	0.11/0.16/0.21 – 0.20/0.26/0.33 (67.9%)	0.11/0.16/0.21 – 0.20/0.26/0.32 (69.7%)	0.10/0.11/0.18 – 0.18/0.21/0.25 (83.7%)
BEV	No	1.10/1.82/3.27 – 4.04/5.05/7.28 (0.0%)	1.02/1.53/2.54 – 4.03/4.73/7.17 (0.0%)	1.02/1.51/2.51 – 4.03/4.72/7.07 (0.0%)	0.97/1.29/2.18 – 4.03/4.72/7.28 (0.0%)
BEV, refined on full	No	0.12/0.19/1.04 – 0.22/0.31/0.83 (49.5%)	0.11/0.17/0.23 – 0.20/0.27/0.38 (64.1%)	0.11/0.16/0.22 – 0.20/0.26/0.36 (65.7%)	0.10/0.12/0.18 – 0.18/0.21/0.26 (79.0%)
Full	Yes	1.18/2.42/4.70 – 0.96/3.22/21.02 (0.1%)	1.09/2.36/5.14 – 1.67/6.45/25.70 (0.1%)	1.08/2.36/5.16 – 1.78/6.80/26.64 (0.1%)	0.95/2.05/3.61 – 2.46/6.76/22.17 (0.1%)
BEV	Yes	1.18/1.86/2.94 – 0.58/1.46/5.79 (0.3%)	1.14/1.75/2.70 – 0.73/2.82/7.28 (0.3%)	1.14/1.75/2.70 – 0.75/3.24/7.32 (0.3%)	1.18/1.79/2.81 – 1.60/4.41/8.14 (0.1%)
BEV, refined on full	Yes	0.15/0.28/2.78 – 0.25/0.42/1.42 (37.8%)	0.12/0.20/2.15 – 0.22/0.31/0.66 (48.8%)	0.12/0.19/2.17 – 0.22/0.31/0.64 (50.1%)	0.11/0.18/3.84 – 0.19/0.24/0.85 (54.5%)

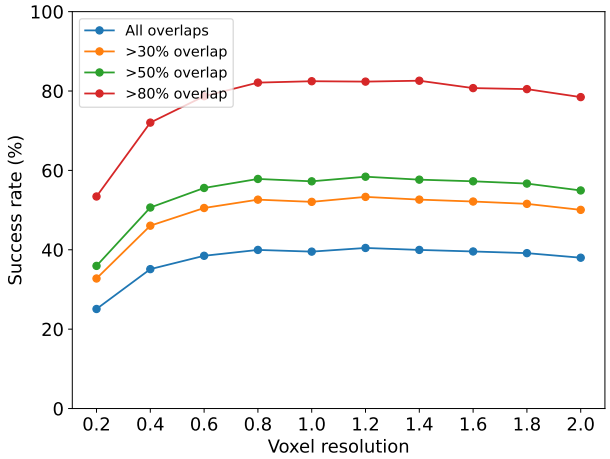


Fig. 15: Success Rate as a function of `voxel_resolution`, swept from 0.2 m to 2.0 m, under the lower bound of the combined planar-and-yaw noise regime.

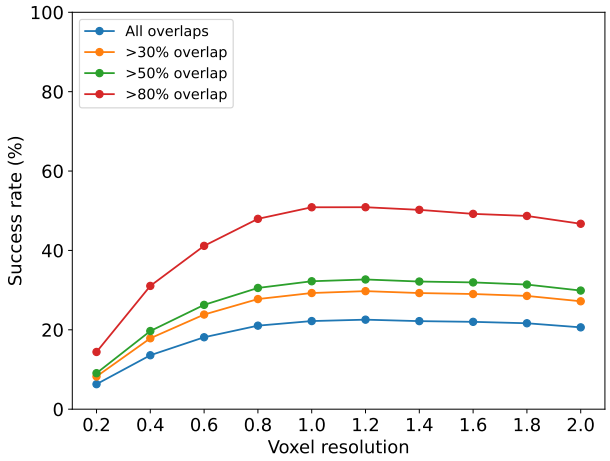


Fig. 16: Success Rate as a function of `voxel_resolution`, swept from 0.2 m to 2.0 m, under the upper bound of the combined planar-and-yaw noise regime.

result with a subsequent registration on the full point clouds: rather than merely failing to improve on the baseline, these also perform worse than it, suggesting that the BEV-derived initialization actively worsens the initial estimate passed to the refinement step.

Ground-plane removal shows a similar pattern when applied to direct registration on the full point clouds, introducing a substantial increase in error that renders the resulting configuration unusable due to its low Success Rate. For

TABLE IV: Mean Runtime ($\pm$ standard deviation) per pipeline configuration.

Point Cloud	Ground removed	Runtime [ms]
Full	No	$104.47 \pm 16.34\,ms$
BEV	No	$60.35 \pm 9.74\,ms$
BEV, refined on full	No	$159.82 \pm 21.84\,ms$
Full	Yes	$863.57 \pm 79.95\,ms$
BEV	Yes	$857.72 \pm 81.76\,ms$
BEV, refined on full	Yes	$947.71 \pm 88.00\,ms$

BEV registration specifically, however, ground-plane removal provides a small improvement over the corresponding non-removal variant, suggesting it partially mitigates the point-intermixing effect discussed above. Despite this partial improvement, all BEV-based configurations, with or without ground-plane removal, still perform worse than the baseline while incurring additional Runtime. Runtime alone further rules out the ground-plane-removal configurations: as shown in Table IV, these require close to one second to execute, making them impractical for any near-real-time application regardless of their accuracy. We therefore discard all six evaluated variations and retain the original registration pipeline (full point clouds, no projection, no ground-plane removal) for the remainder of this work.

D. Beamagine

Since no ground-truth poses are available for this data (Section IV-E), we assess registration quality qualitatively, by visually inspecting the alignment between the vehicle and infrastructure point clouds at different points in the sequence. As a quantitative video or animation cannot be included in this document, we instead show three representative frames from the full 105-frame sequence, illustrating the registration's behavior at its start, midpoint, and near the point where its quality begins to degrade. We use the best-performing configuration identified in Section IV-D, together with the first-order (constant-velocity) sequential initialization strategy introduced in Section V-B, given its improved stability over the zero-order strategy under the upper bound of the combined noise regime.

Figure 17 shows the first frame of the sequence, where the manual initial estimate (Section IV-E) is used directly as the input to VGICP. With this initialization, the algorithm registers both point clouds with an acceptable error and mantains this accuracy for the remaining initial frames.

Figure 18 shows a frame near the middle of the sequence, where sequential initialization (Section III-C) has been propagated from the first frame's estimated transformation across the

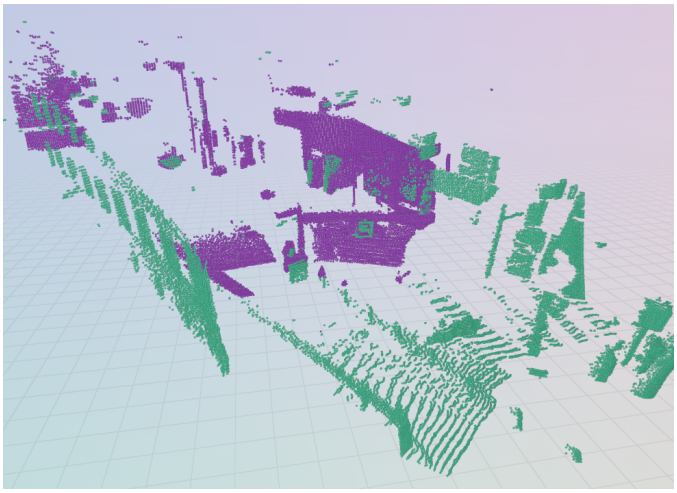


Fig. 17: Registration result for the first frame of the Beamagine sequence, using the manual initial estimate.

intervening frames. The registration remains visibly accurate at this point, with static structures visible to both sensors, such as walls and poles, correctly aligned between the two point clouds.

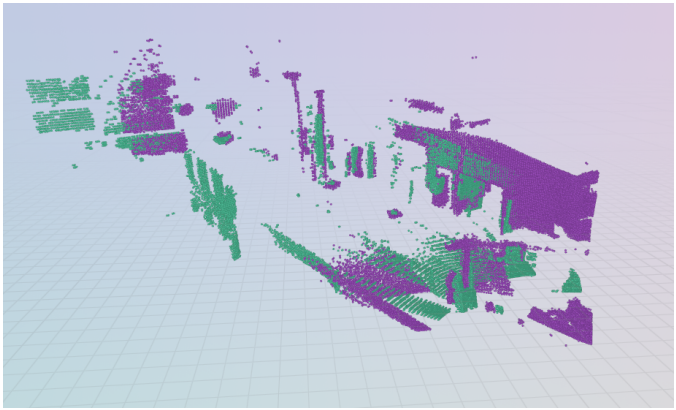


Fig. 18: Registration result for a mid-sequence frame of the Beamagine sequence, under sequential initialization.

Figure 19 shows frame 63 of the sequence, where the registration has visibly deteriorated, coinciding with a reduction in overlap between the vehicle and infrastructure point clouds at this point in the trajectory. This is consistent with the sequential initialization results reported in Section V-B, where Success Rate was observed to decline for frames beyond approximately 50 in a sequence; frame 63 falls within this range, suggesting that the degradation observed here reflects the same underlying limitation identified in that analysis, rather than a failure specific to this sequence.

Taken together, these qualitative results are consistent with the quantitative findings of Section V-B: sequential initialization can sustain accurate registration over a substantial portion of a sequence starting from a single coarse initial estimate, but its reliability declines as the sequence progresses or the overlap diminishes, in agreement with the frame-dependent Success Rate degradation observed under the combined planar-and-yaw

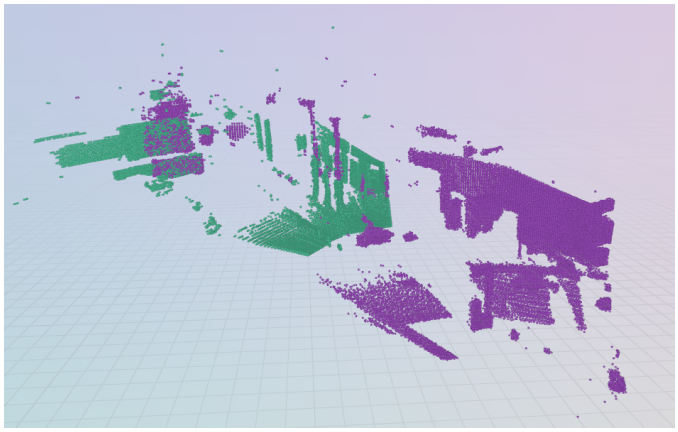


Fig. 19: Registration result for frame 63 of the Beamagine sequence, illustrating the deterioration in registration quality observed toward the later portion of the sequence.

noise regime evaluated earlier in this work.

VI. Conclusions

This work characterizes the applicability of VGICP, a classical, feature-free registration method designed and validated for the scan-to-scan setting, to vehicle-to-infrastructure registration. We evaluate it on V2X-Real in both settings, introduce a controlled noise model reflecting the error characteristics of real GPS/IMU localization, propose a sequential initialization strategy that propagates each frame's estimate from the previous registration rather than from sensor readings, explore the effect of VGICP's parameters on accuracy and Runtime, and validate the resulting configuration on real sensor data collected by Beamagine. We draw the following conclusions with respect to the contributions stated in Section I.

Degradation from scan-to-scan to V2I. VGICP remains usable in the V2I setting, but only within a bounded region around the true transformation. As the comparison in Section V-A makes clear, it fails outright without a reasonable initial estimate, and its accuracy is further conditioned by scene overlap. The limiting factor is therefore the initialization and the geometry of the encounter rather than the registration method itself, and performance measured in the scan-to-scan setting does not predict performance in this one.

Tolerance to initialization error. What counts as a reasonable initial estimate is not a single threshold. Separating the degrees of freedom, as Section V-B does, reveals an error budget that is far from uniform across them: attitude error only slightly reduces Success Rate at realistic magnitudes, while planar displacement and heading dominate. A deployed system should therefore invest in horizontal positioning and heading accuracy rather than in attitude. We also conclude that the rotational error magnitudes assumed in prior work exceed both what the method can absorb and what current GNSS/INS hardware exhibits.

Sequential initialization. The strategy proposed in Section III-C reduces dependence on repeated, individually noisy positioning readings: rather than requiring a fresh sensor-derived estimate at every frame, it requires only a single initial

estimate, propagated across subsequent frames via the previous frame's registered transformation. This reduces error relative to the initial, single-shot estimate over the following frames of a sequence, though accumulated drift makes this a bounded-horizon strategy rather than an open-ended one. This drift can be partially mitigated by extrapolating the vehicle's motion under a constant-velocity assumption rather than treating it as stationary between frames, which stabilizes the declining trend under more severe noise conditions, though it offers little benefit when drift is already small. A single accurate estimate can thus stand in for continuous positioning input over a substantial portion of a sequence, particularly when combined with this more informed motion model.

Parameter and pipeline study. Improving the registration itself proves harder than improving its initialization. Of the configurations compared in Section IV-D, the unmodified pipeline with a tuned voxel resolution is the best we have found. The intuition that discarding height or ground geometry simplifies V2I registration is wrong: both carry constraints the method relies on, and ground removal is in any case disqualified by its Runtime. This objective is met only in part, as a defect in `small_gicp` leaves the joint behavior of `voxel_resolution` and `max_correspondence_distance` uncharacterized.

Validation on real data. The recordings examined in Section V-D confirm that these conclusions hold outside a curated dataset. The fact that the drift horizon reproduces on this data suggests it is a property of the sequential initialization strategy rather than of either dataset.

Three directions remain open for future work. Resolving the `small_gicp` defect would allow `voxel_resolution` and `max_correspondence_distance` to be swept jointly, determining whether the observed plateau is genuine or an artifact of the pinned default. The drift under sequential initialization motivates a mechanism that detects divergence and falls back to a per-frame estimate, or fuses the two, extending the usable horizon well past the roughly 50 frames observed here. Finally, given how strongly accuracy tracks overlap, a lightweight predictor of expected overlap from the agents' approximate relative pose would let a deployed system anticipate when registration is reliable rather than discovering it afterward.

APPENDIX A
FULL ERROR DISTRIBUTIONS WITH OUTLIERS

Throughout this work, outliers in RTE and RRE distributions are defined using the standard $1.5 \times$ IQR criterion and omitted from boxplots in the main text for legibility, as discussed in Section V-A. Table V reports the number and percentage of outliers for each experiment presenting boxplots, and the figures below show the corresponding full error distributions, including outliers, for a representative subset of these experiments.

TABLE V: Outlier percentages (RTE / RRE) per experiment and overlap threshold, using the $1.5 \times$ IQR criterion. Sample sizes (N) per overlap threshold: for scan-to-scan rows, All = 16946, $> 30\%$ = 16946, $> 50\%$ = 16946, $> 80\%$ = 16946; for all other rows, All = 28095, $> 30\%$ = 21370, $> 50\%$ = 19602, $> 80\%$ = 9339.

Condition	All	$> 30\%$	$> 50\%$	$> 80\%$
VGICP Evaluation				
Scan-to-scan, GT init	7.5/4.0	7.5/4.0	7.5/4.0	7.5/4.0
Scan-to-scan, identity init	7.5/3.6	7.5/3.6	7.5/3.6	7.5/3.6
V2I, GT init	12.4/8.6	5.9/9.1	3.9/8.3	1.5/4.2
V2I, identity init	1.4/0.0	4.4/0.0	6.4/0.0	1.3/1.7

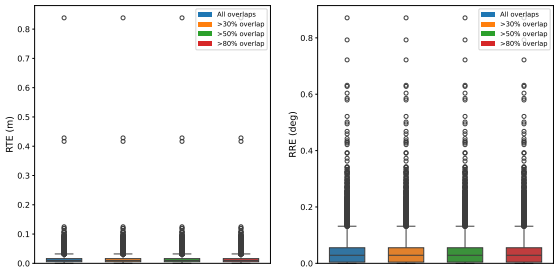


Fig. 20: RTE and RRE for scan-to-scan registration with ground-truth initialization, including outliers.

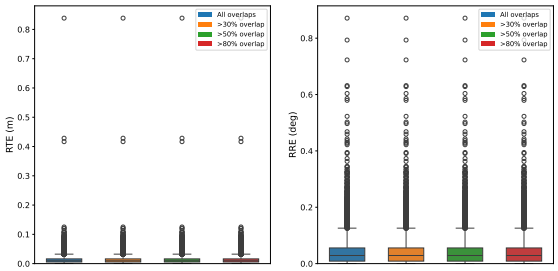


Fig. 21: RTE and RRE for scan-to-scan registration with identity initialization, including outliers.

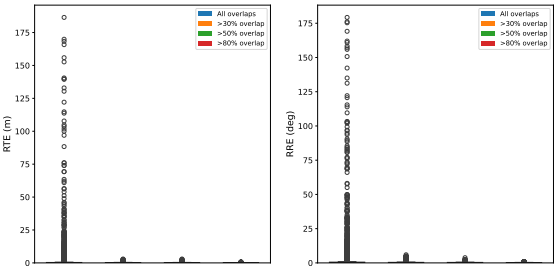


Fig. 22: RTE and RRE for V2I registration with ground-truth initialization, including outliers.

ACKNOWLEDGMENT

This work has been carried out in the context of the 6GEWOC project, which has received funding from the Smart Networks and Services Joint Undertaking (SNS JU) under the European Union's Horizon Europe research and innovation

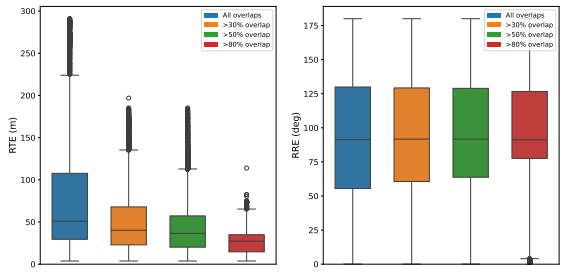


Fig. 23: RTE and RRE for V2I registration with identity initialization, including outliers.

programme under Grant Agreement No. 101139182 (Call: HORIZON-JU-SNS-2023-STREAM-B-01-03).

REFERENCES

[1] M. Seo, H. Lim, K. Lee, L. Carlone, and J. Park, "BUFFER-X: Towards Zero-Shot Point Cloud Registration in Diverse Scenes," pp. 3851–3862, 2025. 1, 2, 6

[2] Z. Zeng, Q. Wu, X. Zhang, L. Y. Wu, P. An, J. Yang, J. Wang, and P. Wang, "Unlocking Generalization Power in LiDAR Point Cloud Registration," pp. 22244–22253, 2025. 1, 2

[3] Y. Chen and G. Medioni, "Object modelling by registration of multiple range images," *Image and Vision Computing*, vol. 10, pp. 145–155, Apr. 1992. 2

[4] A. Segal, D. Haehnel, and S. Thrun, "Generalized-ICP," vol. 05, June 2009. 2, 3, 4

[5] S. Rusinkiewicz, "A symmetric objective function for ICP," *ACM Transactions on Graphics (TOG)*, vol. 38, pp. 85:1–85:7, July 2019. 2

[6] I. Vizzo, T. Guadagnino, B. Mersch, L. Wiesmann, J. Behley, and C. Stachniss, "KISS-ICP: In Defense of Point-to-Point ICP – Simple, Accurate, and Robust Registration If Done the Right Way," *IEEE Robotics and Automation Letters*, vol. 8, pp. 1029–1036, Feb. 2023. 2

[7] K. Koide, M. Yokozuka, S. Oishi, and A. Banno, "Voxelized GICP for Fast and Accurate 3D Point Cloud Registration," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 11054–11059, May 2021. ISSN: 2577-087X. 2, 3, 4, 7

[8] P. J. Besl and N. D. McKay, "Method for registration of 3-D shapes," in *Sensor Fusion IV: Control Paradigms and Data Structures*, vol. 1611, p. 586, SPIE, Apr. 1992. 2

[9] T. Stoyanov, M. Magnusson, and A. J. Lilienthal, "Point set registration through minimization of the L2 distance between 3D-NDT models," in *2012 IEEE International Conference on Robotics and Automation*, pp. 5196–5201, May 2012. ISSN: 1050-4729. 2

[10] H. Hong and B. H. Lee, "Dynamic Scaling Factors of Covariances for Accurate 3D Normal Distributions Transform Registration," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 1190–1196, Oct. 2018. ISSN: 2153-0866. 2

[11] X. Shi, J. Peng, J. Li, P. Yan, and H. Gong, "The Iterative Closest Point Registration Algorithm Based on the Normal Distribution Transformation," *Procedia Computer Science*, vol. 147, pp. 181–190, Jan. 2019. 2

[12] L. Renaut, H. Frei, and A. Nüchter, "Smoothed normal distribution transform for efficient point cloud registration during space rendezvous," in *18th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications, VISIGRAPP 2023*, vol. 5, (Lissabon, Portugal), pp. 919–930, SciTePress, 2023. 2

[13] A. Das, J. Servos, and S. L. Waslander, "3D scan registration using the Normal Distributions Transform with ground segmentation and point cloud clustering," in *2013 IEEE International Conference on Robotics and Automation*, pp. 2207–2212, May 2013. ISSN: 1050-4729. 2

[14] Z. Wang, J. Yi, L. Su, and Y. Pan, "Coherent point drift with Skewed Distribution for accurate point cloud registration," *Computers & Graphics*, vol. 122, p. 103974, Aug. 2024. 2

[15] M. Magnusson, *The Three-Dimensional Normal-Distributions Transform — an Efficient Representation for Registration, Surface Analysis, and Loop Detection*. PhD thesis, Dec. 2009. 2

[16] A. Myronenko and X. Song, "Point Set Registration: Coherent Point Drift," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 32, pp. 2262–2275, Dec. 2010. 2

[17] R. B. Rusu, N. Blodow, and M. Beetz, "Fast Point Feature Histograms (FPFH) for 3D registration," in *2009 IEEE International Conference on Robotics and Automation*, pp. 3212–3217, May 2009. ISSN: 1050-4729. 2

[18] S. Salti, F. Tombari, and L. Di Stefano, "SHOT: Unique signatures of histograms for surface and texture description," *Computer Vision and Image Understanding*, vol. 125, pp. 251–264, Aug. 2014. 2

[19] J. Yang, Q. Zhang, Y. Xiao, and Z. Cao, "TOLDI: An effective and robust approach for 3D local shape description," *Pattern Recognition*, vol. 65, pp. 175–187, May 2017. 2

[20] A. Frome, D. Huber, R. Kolluri, T. Bülow, and J. Malik, "Recognizing Objects in Range Data Using Regional Point Descriptors," in *Computer Vision - ECCV 2004* (T. Pajdla and J. Matas, eds.), (Berlin, Heidelberg), pp. 224–237, Springer, 2004. 2

[21] H. Yang, J. Shi, and L. Carlone, "TEASER: Fast and Certifiable Point Cloud Registration," *IEEE Transactions on Robotics*, vol. 37, pp. 314–333, Apr. 2021. 2

[22] K. Fathian and T. Summers, "CLIPPER+: A Fast Maximal Clique Algorithm for Robust Global Registration," *IEEE Robotics and Automation Letters*, vol. 9, pp. 3562–3569, Apr. 2024. 2

[23] M. Gentner, P. Kumar Murali, and M. Kaboli, "GMCR: Graph-based Maximum Consensus Estimation for Point Cloud Registration," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4967–4974, May 2023. 2

[24] S. Yan, P. Shi, Z. Zhao, K. Wang, K. Cao, J. Wu, and J. Li, "TurboReg: TurboClique for Robust and Efficient Point Cloud Registration," pp. 26371–26381, 2025. 2

[25] Y. Wu, J. Jiang, Y. Yuan, M. Gong, Q. Miao, H. Li, M. Zhang, and W. Ma, "PointTruss: K-Truss for Point Cloud Registration," *Advances in Neural Information Processing Systems*, vol. 38, pp. 51407–51440, Apr. 2026. 2

[26] S. Ao, Q. Hu, H. Wang, K. Xu, and Y. Guo, "BUFFER: Balancing Accuracy, Efficiency, and Generalizability in Point Cloud Registration," pp. 1255–1264, 2023. 2

[27] J. Mu, S. Li, Y. Zhao, and Y. Gao, "ColorPCR++: Unleash the Power of Color for Point Cloud Registration Using Hypergraph Computation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1–16, 2026. 2, 6

[28] Z. Qin, H. Yu, C. Wang, Y. Guo, Y. Peng, S. Ilic, D. Hu, and K. Xu, "GeoTransformer: Fast and Robust Point Cloud Registration With Geometric Transformer," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, pp. 9806–9821, Aug. 2023. 2, 6

[29] W. Lu, G. Wan, Y. Zhou, X. Fu, P. Yuan, and S. Song, "DeepVCP: An End-to-End Deep Neural Network for Point Cloud Registration," pp. 12–21, 2019. 2

[30] Y. Wang, P. Zhou, G. Geng, L. An, and Y. Liu, "CCAG: End-to-End Point Cloud Registration," *IEEE Robotics and Automation Letters*, vol. 9, pp. 435–442, Jan. 2024. 2

[31] A. Kurobe, Y. Sekikawa, K. Ishikawa, and H. Saito, "CorsNet: 3D Point Cloud Registration by Deep Neural Network," *IEEE Robotics and Automation Letters*, vol. 5, pp. 3960–3966, July 2020. 2

[32] V. Sarode, X. Li, H. Goforth, Y. Aoki, R. A. Srivatsan, S. Lucey, and H. Choset, "PCRNet: Point Cloud Registration Network using PointNet Encoding," Nov. 2019. arXiv:1908.07906 [cs.CV]. 2

[33] Y. Aoki, H. Goforth, R. A. Srivatsan, and S. Lucey, "PointNetLK: Robust & Efficient Point Cloud Registration Using PointNet," pp. 7163–7172, 2019. 2

[34] J. Groß, A. Ošep, and B. Leibe, "AlignNet-3D: Fast Point Cloud Registration of Partially Observed Objects," in *2019 International Conference on 3D Vision (3DV)*, pp. 623–632, Sept. 2019. ISSN: 2475-7888. 2

[35] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation," pp. 652–660, 2017. 2

[36] B. D. Lucas and T. Kanade, "An Iterative Image Registration Technique with an Application to Stereo Vision," in *IJCAI'81: 7th international joint conference on Artificial intelligence*, vol. 2, (Vancouver, Canada), pp. 674–679, Aug. 1981. 2

[37] W. Yuan, B. Eckart, K. Kim, V. Jampani, D. Fox, and J. Kautz, "DeepGMR: Learning Latent Gaussian Mixture Models for Registration," in *Computer Vision – ECCV 2020* (A. Vedaldi, H. Bischof, T. Brox,

and J.-M. Frahm, eds.), (Cham), pp. 733–750, Springer International Publishing, 2020. 2

[38] Y. Wu, Y. Yuan, X. Fan, X. Huang, M. Gong, and Q. Miao, "PCRDiffusion: Diffusion Probabilistic Models for Point Cloud Registration," Dec. 2023. arXiv:2312.06063 [cs.CV]. 2

[39] Y. Haitman, A. Efraim, and J. M. Francos, "C-GenReg: Training-Free 3D Point Cloud Registration by Multi-View-Consistent Geometry-to-Image Generation with Probabilistic Modalities Fusion," pp. 3004–3013, 2026. 2

[40] J. Urain, N. Funk, J. Peters, and G. Chalvatzaki, "SE(3)-DiffusionFields: Learning smooth cost functions for joint grasp and motion optimization through diffusion," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 5923–5930, May 2023. 2

[41] R. Xu, H. Xiang, X. Xia, X. Han, J. Li, and J. Ma, "OPV2V: An Open Benchmark Dataset and Fusion Pipeline for Perception with Vehicle-to-Vehicle Communication," in *2022 International Conference on Robotics and Automation (ICRA)*, pp. 2583–2589, May 2022. 3

[42] Y. Li, D. Ma, Z. An, Z. Wang, Y. Zhong, S. Chen, and C. Feng, "V2X-Sim: Multi-Agent Collaborative Perception Dataset and Benchmark for Autonomous Driving," *IEEE Robotics and Automation Letters*, vol. 7, pp. 10914–10921, Oct. 2022. 3

[43] R. Xu, X. Xia, J. Li, H. Li, S. Zhang, Z. Tu, Z. Meng, H. Xiang, X. Dong, R. Song, H. Yu, B. Zhou, and J. Ma, "V2V4Real: A Real-World Large-Scale Dataset for Vehicle-to-Vehicle Cooperative Perception," pp. 13712–13722, 2023. 3

[44] H. Yu, Y. Luo, M. Shu, Y. Huo, Z. Yang, Y. Shi, Z. Guo, H. Li, X. Hu, J. Yuan, and Z. Nie, "DAIR-V2X: A Large-Scale Dataset for Vehicle-Infrastructure Cooperative 3D Object Detection," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, (New Orleans, LA, USA), pp. 21329–21338, IEEE, June 2022. 3

[45] H. Xiang, Z. Zheng, X. Xia, R. Xu, L. Gao, Z. Zhou, X. Han, X. Ji, M. Li, Z. Meng, L. Jin, M. Lei, Z. Ma, Z. He, H. Ma, Y. Yuan, Y. Zhao, and J. Ma, "V2X-Real: A Largs-Scale Dataset for Vehicle-to-Everything Cooperative Perception," in *Computer Vision – ECCV 2024* (A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler, and G. Varol, eds.), (Cham), pp. 455–470, Springer Nature Switzerland, 2025. 3, 5

[46] H. Li, B. Cao, Z. Liang, W. Li, J. Oh, Y. Chen, S. Liang, H. Zhou, C. Ma, J. Liu, Z. Li, P. Zhang, K. Long, M. Liu, J. Jiang, C. Yu, S. Liu, H. Yu, and X. Li, "CATS-V2V: A Real-World Vehicle-to-Vehicle Cooperative Perception Dataset with Complex Adverse Traffic Scenarios," pp. 2294–2303, 2026. 3

[47] K. Koide, "small_gicp: Efficient and parallel algorithms for point cloud registration," *Journal of Open Source Software*, vol. 9, p. 6948, Aug. 2024. 4, 6

[48] X. Zhang, Q. Qu, Y. Xiong, C. Xia, Z. Song, Q. Peng, K. Liu, J. Li, and K. Li, "V2X-Reg++: A Real-Time Global Registration Method for Multi-End Sensing System in Urban Intersections," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, pp. 22913–22925, Dec. 2025. 6, 7, 10

[49] 5G Automotive Association (5GAA), "C-v2x use cases methodology, examples and service level requirements," white paper, 5G Automotive Association, July 2019. 6